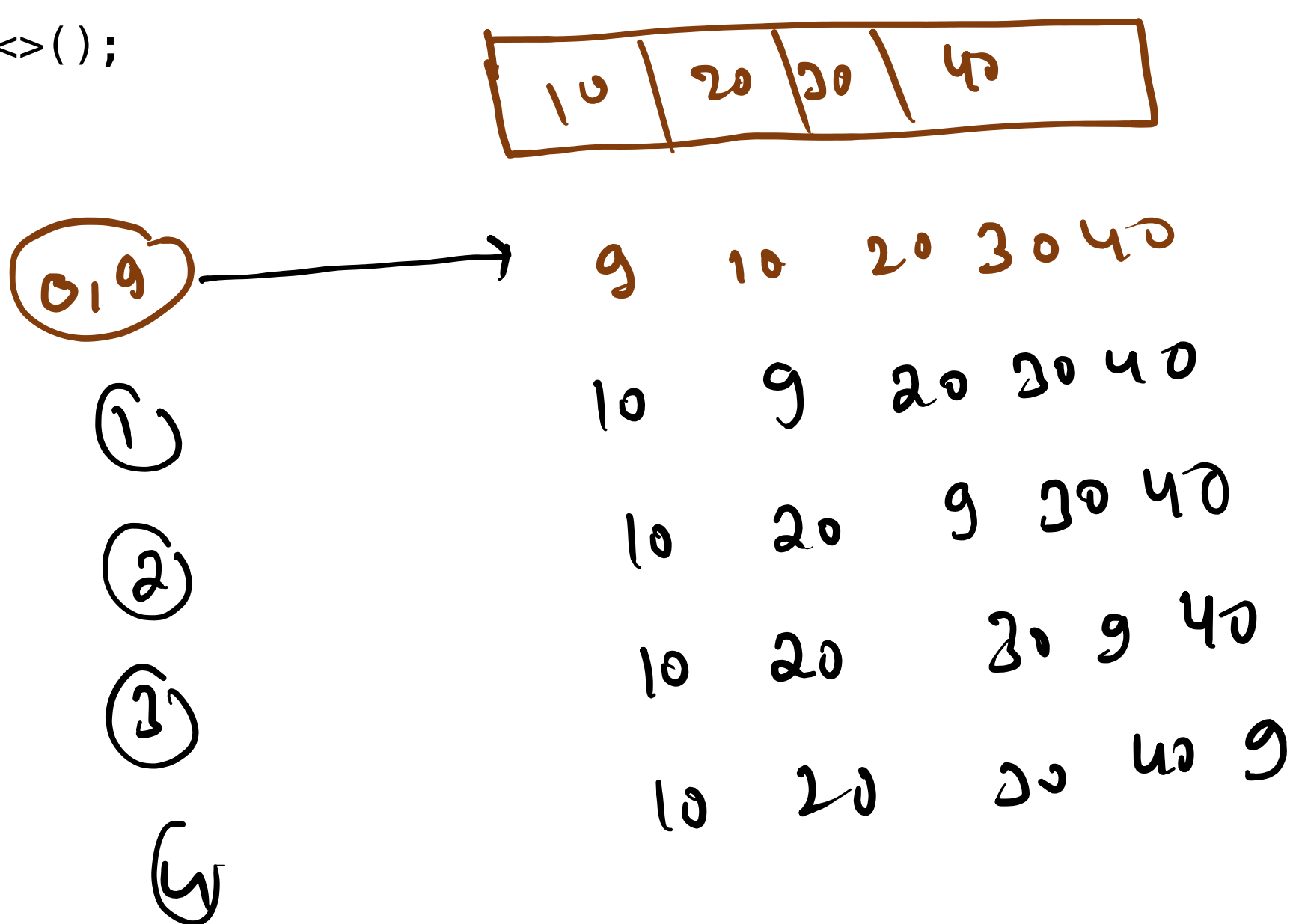


int c) arr = new int(5)

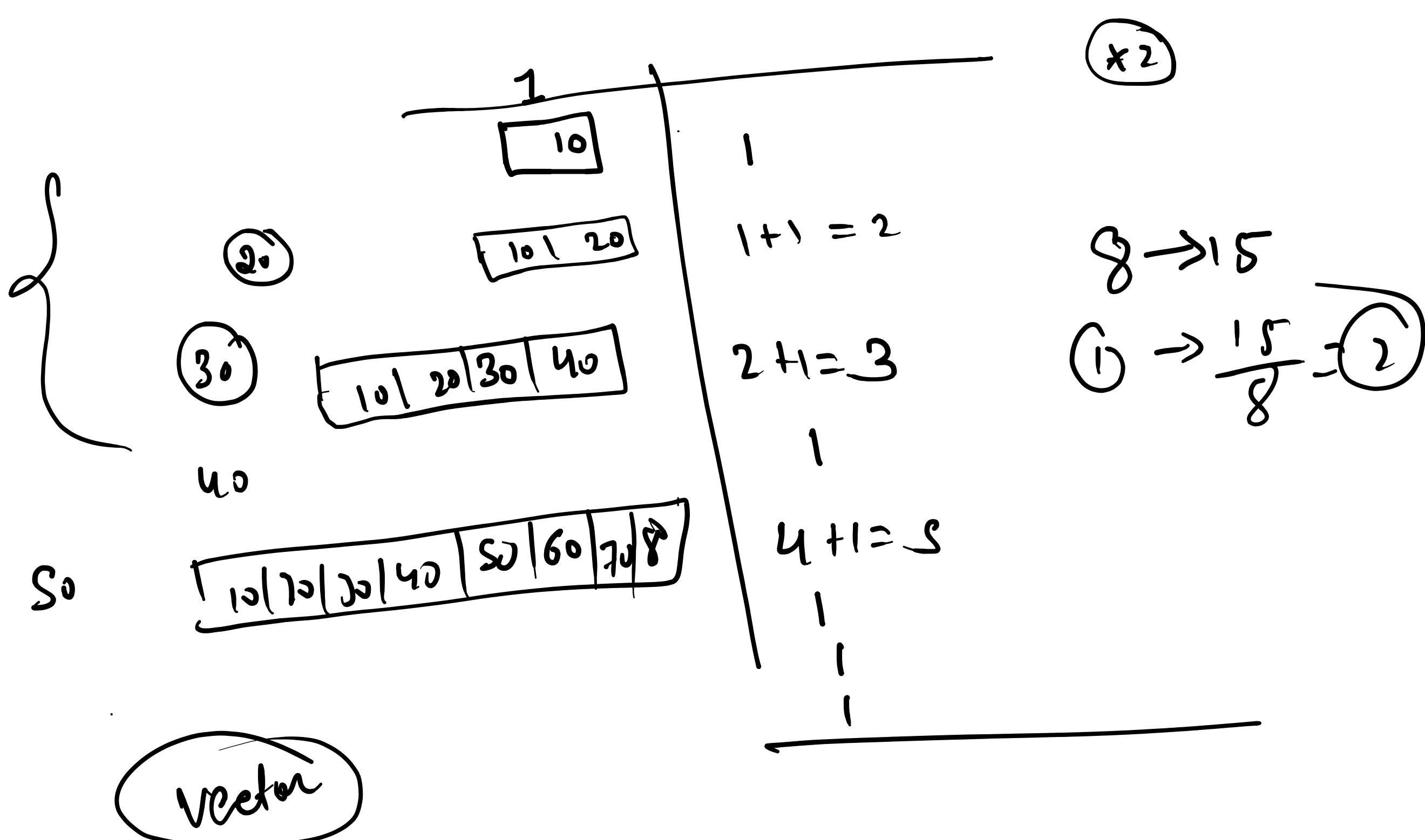
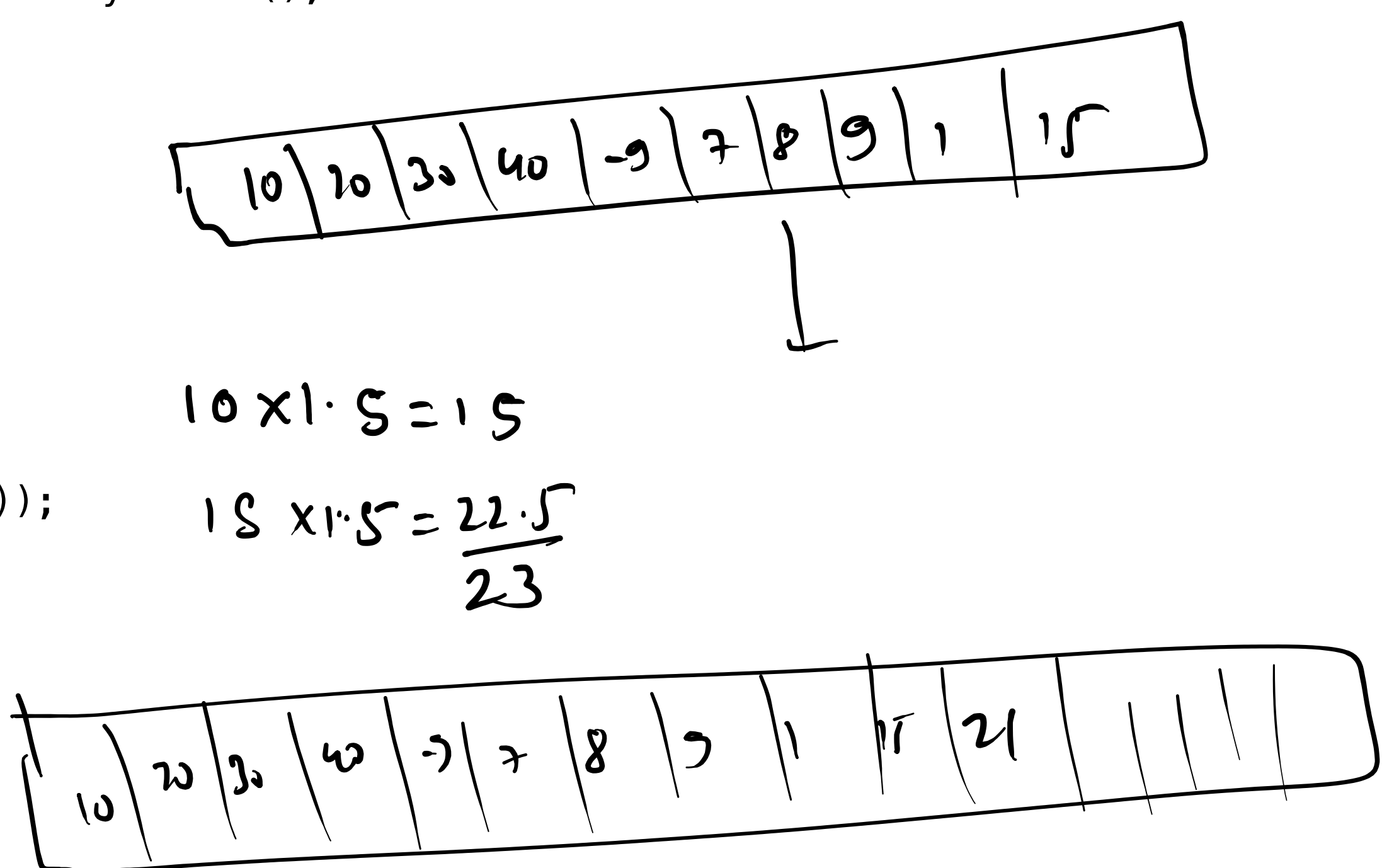
① ArrayList →
↳ class → NP → Heap

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    ArrayList<Integer> ll = new ArrayList<>();  
    // add  
    ll.add(10); // 0(1)  
    ll.add(20);  
    ll.add(30);  
    ll.add(40);  
    System.out.println(ll);  
}
```



```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    ArrayList<Integer> ll = new ArrayList<>();  
    // add  
    ll.add(10);  
    ll.add(20);  
    ll.add(30);  
    ll.add(40);  
    ll.add(-9);  
    ll.add(7);  
    ll.add(8);  
    ll.add(9);  
    ll.add(1);  
    ll.add(15);  
    System.out.println(ll.size());  
}
```

11.02.21



Arrays-Sum Of Two Arrays

Take as input N, the size of array. Take N more inputs and store that in an array. Take as input M, the size of second array and take M more inputs and store that in second array. Write a function that returns the sum of two arrays. Print the value returned.

Example 1

Input

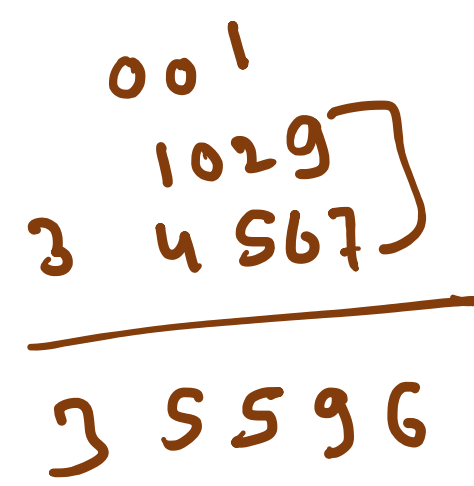
4
1 0 2 9
5
3 4 5 6 7

Output

3, 5, 5, 9, 6, END

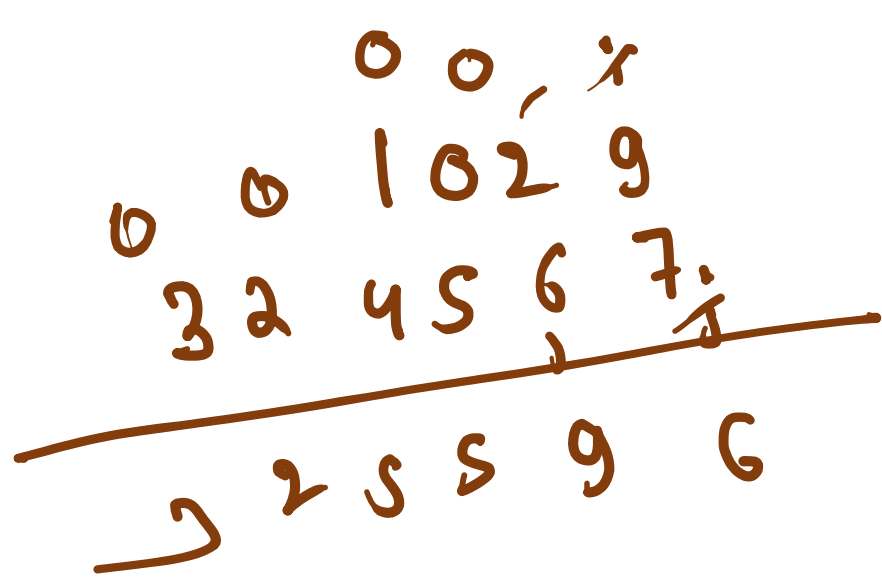
Explanation

Sum of [1, 0, 2, 9] and [3, 4, 5, 6, 7] is [3, 5, 5, 9, 6] and the first digit represents carry over, if any (0 here).

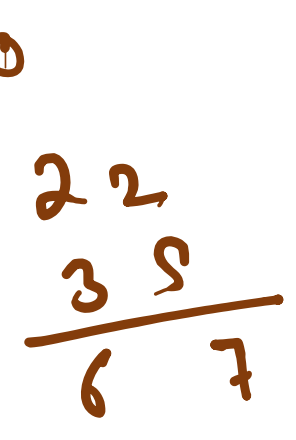
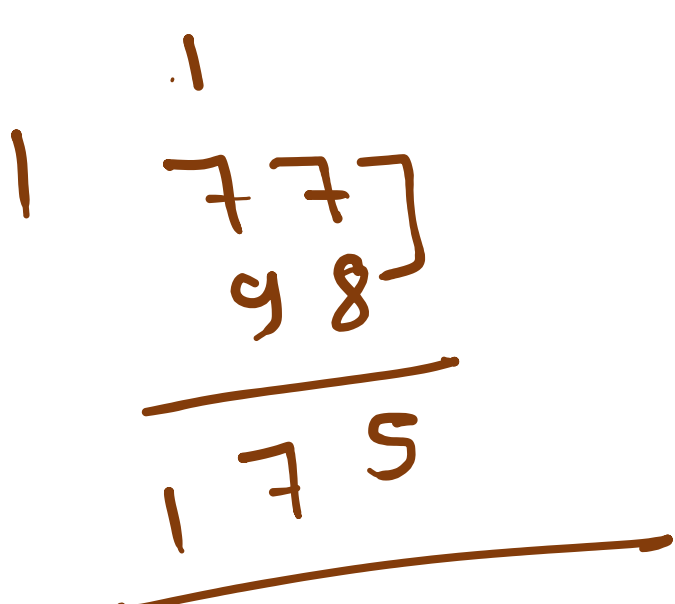


18 div

9/10
carry = 8
carry = 16/10
9/10 = 0



sum = carry + 9 + 7
= 16/10
1 + 2 + 6 = 9

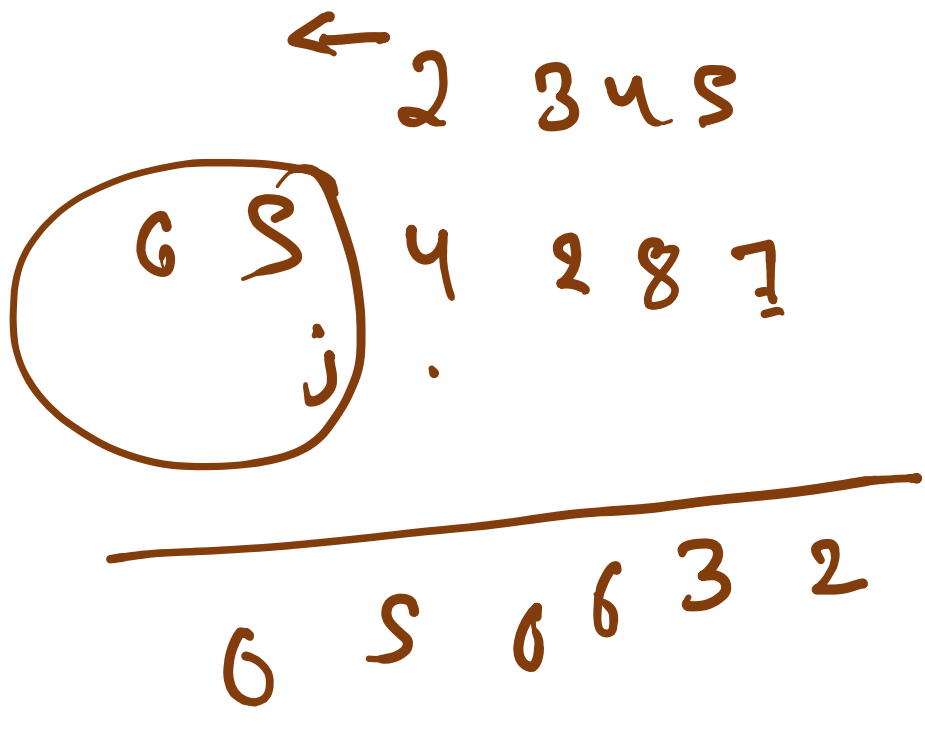


n/m

12



carry = 10



```
public static void Sum_Of_Two_Array(int[] arr1, int[] arr2) {  
    ArrayList<Integer> ll = new ArrayList<>();  
    int i = arr1.length - 1;  
    int j = arr2.length - 1;  
    int carry = 0;  
    while (i >= 0 && j >= 0) {  
        int sum = carry + arr1[i] + arr2[j];  
        ll.add(sum % 10);  
        carry = sum / 10;  
        i--;  
        j--;  
    }  
}
```

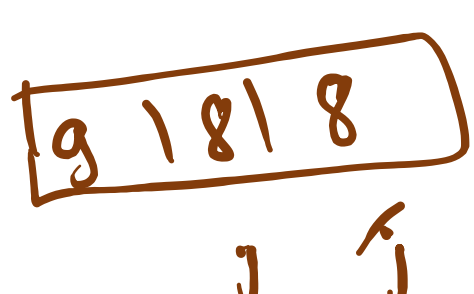
1 + 8 = 9
2 + 3 = 5
3 + 4 = 7
4 + 5 = 9
5 + 6 = 11

carry = 9



```
public static void Sum_Of_Two_Array(int[] arr1, int[] arr2) {  
    ArrayList<Integer> ll = new ArrayList<>();  
    int i = arr1.length - 1;  
    int j = arr2.length - 1;  
    int carry = 0;  
    while (i >= 0 && j >= 0) {  
        int sum = carry + arr1[i] + arr2[j];  
        ll.add(sum % 10);  
        carry = sum / 10;  
        i--;  
        j--;  
    }  
    while (i >= 0) {  
        int sum = carry + arr1[i];  
        ll.add(sum % 10);  
        carry = sum / 10;  
        i--;  
    }  
    while (j >= 0) {  
        int sum = carry + arr2[j];  
        ll.add(sum % 10);  
        carry = sum / 10;  
        j--;  
    }  
}
```

10 - i + 1
7 + 8 = 15



carry = 1
= 0 + 8 + 18
= 1 + 7 + 8
1 + 9 = 10

